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Dear Editor,

Please find enclosed the manuscript **"Where Should Resident Layers Go? Optimal Placement for Weight Streaming in Offloaded Large Language Model Inference"** for consideration as a Research Article in *IEEE Access*.

Problem. Large language models that exceed GPU memory are served by keeping some layers resident and streaming the others from host-attached memory (page-locked DRAM or a CXL memory expander) through a ring of GPU buffers. Widely used tools fill GPU memory in model order, so the resident layers form a contiguous block. We ask where the resident layers should be placed and how many buffers are needed.

Contributions. (1) A max-plus model of the streaming pipeline whose iteration time is the maximum mean of a family of tilings of the layer sequence, with proofs. (2) For layers of equal cost: resident layers are useless when a layer computes at least as long as it copies; evenly spread resident layers attain the lower bound whenever $(k-1)f \geq 2c$; and a contiguous block takes $\max(Lc, mf, rc + 2c + (m-k)f)$, up to $2 - c/f$ times the optimum. (3) A GPU runtime and 605 measured configurations on a Tesla T4 with real layer kernels of two model shapes: given per-layer times measured on the same GPU, the model reproduces the iteration time with a mean error of 0.29%, even spreading is up to 1.50× faster than the contiguous block, and it reaches the bound with two buffers where the contiguous block needs at least five. (4) An honest account of the limits: the power-limited T4 requires clock locking for the quantitative results, and at default clocks the model still ranks placements correctly but overstates their difference.

Fit with the journal. The work addresses the performance of AI inference systems that combine GPU memory with host-attached and CXL memory, combining a performance model with proofs and measurements on real hardware, which fits the computing and systems scope of IEEE Access. Code, raw measurements and the scripts that generate every number and figure are publicly available.

The manuscript is original, is not under consideration elsewhere, and all authors have approved the submission. The authors declare no competing interests. The use of generative artificial intelligence tools is disclosed in the Acknowledgment section, as required by IEEE.

Sincerely,

Quang-Vinh Dang (corresponding author), on behalf of Dat Le and Minh Ngoc Dinh
British University Vietnam, Hung Yen, Vietnam
vinh.dq4@buv.edu.vn ORCID 0000-0002-3877-8024